

Hazard-Aware Urban Navigation Under Flooding: A Review of Routing, Uncertain Crowd Evidence, and Verified Explanation for Offline-First Systems

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Abstract—Urban floods cut road networks in minutes, yet mainstream navigation still optimises travel time on a network assumed to be intact, and it usually stops working when the cellular network fails. This review examines the research needed for a navigation system that routes around flood hazards, states how confident it is, explains its decisions in plain language, and keeps doing all three on a phone without connectivity. We surveyed 129 works published between 1927 and 2025 and verified the bibliographic record of each against Crossref, arXiv, or, for four items, a secondary listing. We organise the literature into eight strands: flood impacts on road networks; hazard-aware and risk-averse route planning; crowdsourced and multi-source hazard evidence; belief, trust, temporal decay and calibration; natural-language explanation and its faithfulness; on-device language models; offline-first mobile systems; and human reliance on uncertain automation. A cross-strand synthesis shows that each capability is well studied in isolation, but that we found no study combining uncertainty-aware flood routing with explanations that are checked against the routing decision and produced fully on-device. We also find that several ideas often presented as new, such as flood-aware routing, decay of report trust with age, and keeping landmark-based search correct when edge weights only increase, have clear precedents. The review closes with eight open problems, including the absence of street-level flood-depth ground truth, correlated crowd reports, calibration under distribution shift between flood events, and measurement of explanation faithfulness on low-cost phones, and it uses Chennai, India, as a running case.

Index Terms—Crowdsourcing, explainable artificial intelligence, flood hazards, mobile computing, route planning, uncertainty.

I. INTRODUCTION

FLOODING is one of the most common ways in which cities lose their road networks. Hydrodynamic studies of Shanghai [1] and York [2], and a city-scale study of emergency responders [3], show that even moderate pluvial events disconnect parts of a city and lengthen emergency response. At regional scale, analyses of the San Francisco Bay Area [4], Boston [5], and accessibility patterns under extreme floods [6] trace how local inundation propagates into system-wide delay, and a study of a city in southern China reports that more than 40% of road connectivity in its study area is lost under a 1-in-100-year rainfall [7]. Vehicle speed also falls well before a road becomes impassable; the depth-disruption

function of Pregnotato *et al.* [8] relates standing-water depth to safe driving speed and has since been used in preparedness studies [9].

Chennai, India, illustrates the problem sharply. The December 2015 event was examined in attribution [10], geospatial [11], and geo-anthropogenic [12] studies; urban sprawl has been linked to rising flood exposure [13]; and a comparison of the 2015 and 2023 floods combines synthetic-aperture radar with reservoir-release information [14]. A national flood inventory places such events in a longer record [15]. What these studies do not provide is live, street-level information that a driver or an ambulance could use while the water is rising.

A navigation system for such conditions needs four capabilities at once. It must *route* on a road graph whose costs depend on uncertain hazard evidence. It must maintain a *belief* about each road that fuses a static susceptibility prior with time-stamped reports of varying reliability, and whose confidence is calibrated. It must *explain* why a detour is proposed without inventing facts. And it must do all of this *offline*, because floods also damage the communication infrastructure on which cloud services depend.

Existing surveys each cover one of these capabilities: route planning in transportation networks [16], evacuation optimisation [17], social media in emergencies [18], the quality of volunteered geographic information (VGI) [19], truth discovery [20], hallucination in natural language generation (NLG) [21], data-to-text generation [22], explainable artificial intelligence (XAI) [23]–[25], and on-device language models [26]. This review reads across them. Its contributions are:

- a taxonomy of eight research strands relevant to offline, uncertainty-aware, explainable flood navigation, with a synthesis of what each strand offers and leaves open (Sections IV–XI);
- a cross-strand comparison (Table I) that separates capabilities with strong precedent from gaps that remain open;
- an assessment of which commonly claimed novelties have prior art, including patents and dynamic-graph routing results (Section XII);
- a list of open problems grounded in the Chennai setting (Section XIII).

II. REVIEW METHODOLOGY

We started from the source lists of seven internal research reports prepared for the CityPulse AI project and extended

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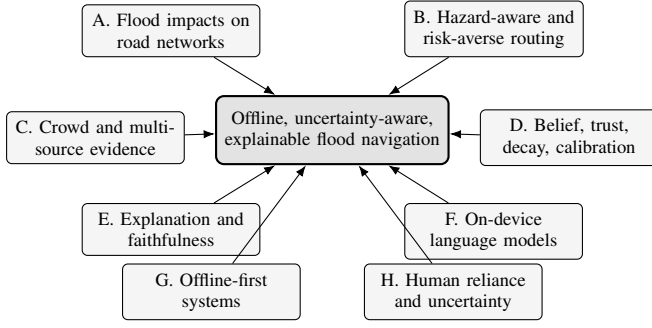


Fig. 1. Taxonomy used in this review. Each strand contributes one capability to the target system in the centre.

them with targeted searches on the themes of Fig. 1. Search strings combined the terms *flood*, *road network*, *routing*, *shortest path*, *stochastic*, *reliable*, *crowdsourced*, *Waze*, *social media*, *volunteered geographic information*, *truth discovery*, *reputation*, *calibration*, *explanation*, *faithfulness*, *hallucination*, *data-to-text*, *on-device*, *quantization*, *local-first*, *CRDT*, *trust in automation*, and *Chennai*. Candidates were retained if they (i) reported a method, measurement, dataset or system relevant to at least one of the eight strands, and (ii) had a bibliographic record we could open.

Every retained item was checked against a primary metadata record in the same session: 92 against Crossref by DOI, 33 against their arXiv abstract pages, and four against secondary listings (one conference paper without a DOI, one ACM Digital Library listing, and two patent records). Twelve further candidates could not be confirmed and were excluded rather than cited from memory: ten because the metadata service rate-limited the lookup, and two because the DOI we held resolved to an unrelated paper. The final corpus contains 129 works published between 1927 and 2025. Where only a title or abstract was available to us, this review describes the work at that level and no further. Because the corpus was assembled by targeted search rather than exhaustive database screening, the review is structured but not a systematic review in the PRISMA sense.

III. BACKGROUND AND TERMINOLOGY

We model a city as a directed graph $G = (V, E)$ whose edges carry a free-flow traversal time $\tau_0(e)$. A *hazard belief* $p(e, t)$ is the probability that edge e is affected by a hazard class (for example, flooding) at time t . *Evidence* consists of timestamped reports with a location, a polarity (present or cleared) and a source whose reliability may be known or estimated. A *static prior* expresses susceptibility before any report arrives; height above the nearest drainage (HAND) [27] and historical inundation maps are typical inputs. *Calibration* means that, among edges assigned probability q , a fraction close to q are truly affected; it is distinct from *discrimination*, which asks only whether affected edges receive higher scores [28]. An explanation is *faithful* if it reflects the reasons the system actually used [29]; in a closed domain such as routing, a stricter and checkable requirement is that every factual token in the explanation is grounded in the router’s decision record.

IV. FLOOD IMPACTS ON ROAD NETWORKS

The first strand quantifies how floods degrade mobility. Coupled hydrodynamic and network models dominate. Yin *et al.* [1] assess pluvial flash-flood risk to an intra-urban road network; Coles *et al.* [2] and Green *et al.* [3] model how far emergency responders can reach during floods, moving the question from “which streets flood” to “which people lose service”. Zhou *et al.* [7] combine one-dimensional drainage modelling with two-dimensional inundation mapping and report connectivity loss as rainfall intensity grows. System-level studies of Boston [5] and the Bay Area [4] show that disruption spreads well beyond the flooded links, and accessibility-based vulnerability patterns have been mapped for extreme floods [6].

Two results from this strand matter for navigation. First, disruption is gradual: the depth–disruption function [8] lets a router slow an edge rather than delete it, and preparedness analyses build on it [9]. Second, the models are scenario tools. They are run for design storms, often with high-resolution terrain and drainage data that most cities, Chennai included, do not publish. Chennai-specific work is mainly retrospective: rainfall attribution for 2015 [10], multi-criteria GIS analysis [11], anthropogenic drivers [12], land-use projections [13], and a 2015 versus 2023 comparison [14]. Global forecasting services have improved sharply, from ensemble streamflow forecasting [30] to machine-learning prediction in ungauged basins [31], but they forecast rivers and catchments rather than individual streets.

What remains: none of these works turns into a street-level, time-stamped passability estimate that can be updated from reports during an event.

V. HAZARD-AWARE AND RISK-AVERSE ROUTE PLANNING

A. Shortest paths and speed-up techniques

Dijkstra’s algorithm [32] and A* search [33] remain the basis of most routers. Goal-directed search with landmarks and the triangle inequality (ALT) [34], contraction hierarchies [35], and their customisable variants [36], [37] make continental queries fast; Bast *et al.* [16] survey the field, and time-dependent variants store shortcut paths rather than travel-time functions [38]. Open-source engines such as OSRM brought these techniques to OpenStreetMap data [39], [40].

Changing weights are the central difficulty for hazard routing, because every new report changes costs. Customisable approaches separate metric-independent preprocessing from a cheap customisation step [36], [37]. For landmark-based search, Delling and Wagner showed that ALT “still performs correct queries as long as an edge weight does not drop below its initial value” [41]. This result matters for any hazard cost that never falls below free-flow time: landmark potentials computed once on free-flow times stay valid for every hazard configuration. The property should therefore be credited to the dynamic-graph literature rather than presented as new to hazard routing.

B. Stochastic, reliable and robust routing

When edge costs are uncertain, the objective itself changes. Early work defined shortest paths in probabilistic graphs [42]; later studies addressed expected and least-expected-time paths in dynamic stochastic networks [43], [44], on-time arrival probability [45], and tractable reliable routing [46]. Chance-constrained programming [47] and robust discrete optimisation [48] provide the two classical ways to be cautious: bound the probability of violating a constraint, or optimise against a set of adversarial costs. Hazardous-materials routing has long weighed travel cost against exposure risk [49].

C. Flood-aware routing

Several recent studies route directly on flood models: risk-avoidance routes planned with Dijkstra’s algorithm over a high-resolution numerical flood model [50], shortest-path planning combined with dynamic dispatch of rescue forces [51], deep reinforcement learning for rescue path planning [52], and evacuation routing for dam-failure floods on a coupled flood–road model [53]. Evacuation research adds capacity constraints [17], [54]. On the industrial side, a flood-aware travel-routing application [55] and a patent on routing with per-segment safety scores [56] show that the basic idea of hazard-aware routing is not new.

What remains: most flood-routing studies assume the flood state is known (from a model) and deterministic. Few treat hazard evidence as sparse, stale and unreliable, and fewer adjust caution according to how much evidence exists.

VI. CROWDSOURCED AND MULTI-SOURCE HAZARD EVIDENCE

Goodchild’s view of “citizens as sensors” [57] and its application to disaster response [58] framed a large literature on VGI. OpenStreetMap itself is the best-known example [40], and quality-assessment methods for VGI have been reviewed [19]. Social media added a real-time channel: tweets have been used to detect earthquakes [59], and a survey catalogues methods to filter, classify and summarise emergency messages [18]. Combining social media with authoritative data improves the identification of useful messages [60].

For floods specifically, crowdsourced and social-media observations have been used to monitor urban flooding at high resolution [61], to map inundation rapidly with remote sensing and topography [62], to support flood modelling more generally [63], and to model coastal flood severity [64]. Navigation-app data is especially relevant. Waze incident reports have been assessed for reliability, coverage and added value [65]; flood reports from Waze have been checked for trustworthiness in Norfolk, Virginia [66]; and in Dallas, Waze flood reports were found to align more closely with surface depressions than with conventional flood hazard maps, with an empirical Bayes model outperforming random-forest regression [67].

Truth discovery provides principled tools for conflicting reports. The Dawid–Skene model estimates observer error rates by expectation–maximisation [68]; later methods estimate source reliability jointly with the truth [69], [70], as surveyed by Li *et al.* [20].

What remains: crowd reports cluster in space and time and are often duplicates, yet many fusion schemes treat them as independent. Negative evidence (a road that people drove through without reporting a problem) is rarely modelled, although it is exactly what a navigation app observes most often.

VII. BELIEF, TRUST, TEMPORAL DECAY AND CALIBRATION

A. Fusing evidence over time

Occupancy grids established probabilistic fusion of noisy sensor readings for robot navigation [71]; they are now commonly implemented as additive log-odds updates. Robotics later addressed features that disappear: the persistence filter of Rosen *et al.* [72] estimates whether a previously observed feature still exists, which is the same question a navigation app faces for a reported flood. In reputation systems, the beta reputation system with forgetting [73], subjective logic [74], and broader trust surveys [75] formalise reliability and its decay. Decay of trust with time is therefore well established.

B. Pessimism and uncertainty bounds

Upper confidence bounds are familiar from bandit algorithms [76], where optimism drives exploration. Safety-critical routing needs the opposite: pessimism when evidence is thin. The simplest normal-approximation (Wald) interval for a proportion behaves poorly near zero and one, and the Wilson score interval [77], the adjusted interval of Agresti and Coull [78], and the analysis of Brown *et al.* [79] give better-behaved alternatives. Any pessimistic plug-in used by a router inherits these properties.

C. Evaluating probabilities

The Brier score [80] and Murphy’s decomposition into reliability, resolution and uncertainty [81], together with the forecaster comparison framework of DeGroot and Fienberg [82] and strictly proper scoring rules [83], remain the standard tools. Machine learning revived interest in calibration [84]–[86] and in the pitfalls of expected calibration error [87]. Receiver operating characteristic analysis measures discrimination separately [28].

What remains: hazard-class-specific decay constants are seldom fitted to real closure durations, calibration is rarely reported for hazard beliefs, and evaluations seldom split data by flood event, although reports from a single event are strongly correlated.

VIII. EXPLANATION AND FAITHFULNESS

A. What makes an explanation useful

Miller argues from the social sciences that people want contrastive, selective explanations [88]. Position papers call for rigorous evaluation of interpretability [89], [90], and metrics for XAI have been proposed [91]. Surveys organise explanation methods for black-box models [23]–[25]. In driving, Koo *et al.* [92] studied explanations of semi-autonomous driving actions as a means to improve driver understanding, trust and performance.

B. Generating explanations from structured data

Classical NLG pipelines [93] and the observation that template-based and “real” NLG are not opposites [94] suggest that closed-domain explanations can be rendered reliably from structured records. Data-to-text benchmarks exposed hallucination and content-selection problems in neural systems [22], [95].

C. Faithfulness and hallucination

Hallucination is now a central NLG problem [21]. Chain-of-thought explanations can misrepresent why a model answered as it did [96], and faithfulness needs its own definition and evaluation [29]. Factual-consistency checkers [97], meta-evaluations of such metrics [98], atomic-fact scoring [99], sampling-based self-consistency checks [100], retrieval augmentation [101], citation generation [102] and constrained decoding [103] all reduce or detect unsupported content.

What remains: these methods target open-domain text and estimate faithfulness statistically. A routing explanation lives in a closed domain in which every admissible fact is known in advance, so exact symbolic verification against the decision record is possible, yet we found no study that evaluates such verification for navigation explanations.

IX. ON-DEVICE LANGUAGE MODELS

Quantisation made local inference practical: integer-only inference for mobile networks [104], 8-bit matrix multiplication [105], post-training quantisation [106], activation-aware weight quantisation [107], and parameter-efficient finetuning [108], [109]. Small open models target phones directly [110]–[113]. Measurements on commercial phones report large differences in speed, energy and memory across platforms [114], [115], and PalmBench also documents hallucinated and toxic output from compressed models [115]. Scaling-law evidence suggests that low-bit quantisation degrades well-trained small models more than undertrained ones [116]. A broad review covers architectures, compression and edge–cloud collaboration [26].

What remains: almost all measurements come from flagship or developer hardware, and none that we found measure a safety-critical, grounded generation task under the thermal and memory limits of low-cost phones.

X. OFFLINE-FIRST MOBILE SYSTEMS

Mobile systems research has long treated disconnection as normal. Delay-tolerant networking designs for challenged networks [117], logical clocks [118] and hybrid logical clocks [119] order events without reliable connectivity, and conflict-free replicated data types (CRDTs) [120] let replicas merge without coordination. The local-first software agenda argues that data should live primarily on the user’s device [121]. Spatial indexing on the device, for example with R*-trees [122], keeps hazard lookups fast without a server.

What remains: the offline-first literature is domain-agnostic; it does not address how a safety belief should age while a device is disconnected, or how a returning device should reconcile its local decisions with newer shared evidence.

XI. HUMAN RELIANCE ON UNCERTAIN AUTOMATION

Trust should match reliability; Lee and See frame this as appropriate reliance [123], building on the categories of use, misuse, disuse and abuse [124]. A widely used trust scale was derived empirically by Jian *et al.* [125]. Showing model confidence can help people calibrate their trust, but calibration alone did not improve joint decisions in the experiments of Zhang *et al.* [126], and explanations sometimes increased acceptance of AI recommendations regardless of correctness [127]. Uncertainty displays for everyday mobile predictions [128] and qualitative expressions of forecaster confidence [129] show that presentation changes decisions.

What remains: a navigation app that says “no recent data” for a corridor behaves differently from one that silently treats the corridor as clear, but we found no study measuring appropriate reliance on hazard-routing advice when the system’s belief is sometimes wrong.

XII. CROSS-STRAND SYNTHESIS

Table I summarises the maturity of each strand and what it leaves open. Three observations follow.

First, the component ideas are mature, and several popular claims of novelty have precedent. Flood-aware routing exists in academic work [50]–[53] and in patents [55], [56]. Decay of report trust has long roots in reputation systems [73], [75] and robotics [72]. Correctness of landmark search under weight increases is established [41]. Crowdsourced flood detection from navigation apps has been studied [66], [67].

Second, the combination is not addressed. We found no work that simultaneously (i) routes on beliefs that fuse a static prior with sparse, decaying crowd evidence, (ii) becomes more cautious when evidence is thin rather than less, (iii) reports calibrated confidence, (iv) renders explanations that are symbolically verified against the routing decision, and (v) runs entirely on a phone. We state this as the limit of our search, not as proof of absence.

Third, evaluation practice lags behind system design. Flood-routing studies typically evaluate against the same model that produced the flood map, so the router cannot be wrong about the hazard. Honest evaluation needs ground truth that is independent of the belief, splits by event, and metrics that do not reward a configuration for being scored under its own assumptions.

XIII. OPEN PROBLEMS

- 1) *Ground truth for street-level passability.* Retrospective Chennai studies [11], [14] and national inventories [15] describe events, not time-stamped street closures. Cameras [61], navigation-app reports [66], [67] and field campaigns are candidate sources, each with coverage bias.
- 2) *Depth, not just presence.* Depth–disruption models [8] need depth estimates, but most crowd reports carry only presence. Categorical depth reports and image-based estimates are open design questions.
- 3) *Correlated and adversarial reports.* Truth-discovery methods [68]–[70] estimate reliability but rarely model

TABLE I
MATURITY OF THE EIGHT STRANDS AND THE GAP EACH LEAVES FOR OFFLINE, UNCERTAINTY-AWARE, EXPLAINABLE FLOOD NAVIGATION

Strand	Representative works	Established capability	Gap relevant to flood navigation
A. Flood impacts	[1]–[4], [7], [8]	Depth–speed relations; accessibility loss under design storms	No live, street-level passability estimate
B. Routing	[34], [36], [41], [46], [50], [51]	Fast queries under changing weights; reliable and flood-aware paths	Hazard state usually assumed known and deterministic
C. Crowd evidence	[18], [20], [57], [61], [66], [67]	Crowd and social data detect floods; source reliability estimation	Correlated duplicates; negative evidence rarely used
D. Belief and calibration	[71]–[73], [80], [81], [86]	Log-odds fusion; persistence; proper scores	Class-specific decay rarely fitted; event-split calibration rare
E. Explanation	[21], [29], [88], [94], [98], [99]	Contrastive explanation theory; hallucination detection	Exact verification against a routing decision record untested
F. On-device models	[106], [107], [111], [113]–[115]	Sub-billion to few-billion parameter models run on phones	Little evidence on low-cost phones for grounded tasks
G. Offline-first	[117], [119]–[122]	Disconnected operation and conflict-free merging	Ageing of safety beliefs while disconnected
H. Human reliance	[123]–[127], [129]	Appropriate reliance; confidence displays	No reliance study for hazard-routing advice

duplication by the same event or deliberate false reports in a safety setting.

- 4) *Negative evidence under selection bias.* Traversal without a report is evidence of passability only if drivers did not avoid the road because they expected flooding; the missing-not-at-random structure must be modelled.
- 5) *Calibration under shift between events.* Proper scores [83] and reliability diagrams [82] should be reported on held-out events, because monsoons differ in timing, reservoir releases [14] and urban form [13].
- 6) *Cost models that capture harm.* Proportional penalties inflate the cost of long exposure but can leave short, severe segments such as flooded underpasses cheap; chance constraints [47] and robust formulations [48] offer alternatives whose behaviour on real networks needs measurement.
- 7) *Verified explanation on low-cost phones.* Exact verification of closed-domain explanations and its pass rate on phones typical of the target population remain unmeasured [114], [115].
- 8) *Appropriate reliance.* Studies must include trials in which the system is wrong, measuring reliance when the advice is correct against reliance when it is not [123], [127].

XIV. CONCLUSION

The research needed for flood-resilient navigation exists in eight separate literatures. Routing algorithms already handle changing weights, crowd and navigation-app data already detect floods, belief and calibration methods are mature, closed-domain generation can be checked exactly, small models run on phones, and offline-first systems handle disconnection. The unsolved work lies in combining them under honest evaluation: ground truth independent of the belief, splits by flood event, cost models that reflect harm, and verification that works on the phones people in flood-prone neighbourhoods actually own. For Chennai, where major floods recurred in 2015 and 2023 [14] and connectivity fails when it is most needed, this combination is the practical research agenda.

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